

The Role of Multi-Agents in Digital Twin Implementation: Short Survey

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In recent years, Digital Twin (DT) technology has emerged as a significant technological advancement. A digital twin is a digital representation of a physical asset that mirrors its data model, behaviour, and interactions with other physical assets. Digital Twin aims at achieving adaptability, seamless data integration, modelling, simulation, automation, and real-time data management. The primary goal of this article is to explore the role of agents in DT implementations, seeking to understand their predominant usage scenarios and purposes. From our perspective, agents serving as intelligent entities play a role in realising the features of DTs. This article also discusses the gaps in DT, highlights future directions, and analyses various technologies integrated with multi-agent systems technologies in DT implementations. Finally, the article briefly discusses an overview of an architecture to implement a DT for smart agriculture with multi-agents.

CCS Concepts: • **Computing methodologies** → **Multi-agent systems**; • **General and reference** → **Surveys and overviews;**

Additional Key Words and Phrases: Agents, digital twin, multi-agent systems

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1 Introduction

Digital Twins (**DT**) have been getting more attention in the last decade as an essential technology in the rapidly evolving cyber-physical systems. By creating a digital counterpart of physical entities in a digital environment, DTs enable a deeper understanding of systems, improve predictive maintenance, and introduce features like adaptability [49]. Various domains, including manufacturing [42], healthcare [47], smart agriculture [19], smart cities [9], and others, have already started utilising power of Digital Twin. With the increased attention of DTs in recent years, the exploration of Digital Twins alongside many technologies has increased. Among these, **Multi-Agent Systems** (**MAS**) play an important role due to the unique capabilities of agents, moving DTs towards more autonomy, efficiency, and cognitive intelligence.

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In computational and artificial intelligence systems, the concept of an "agent" refers to a software entity capable of acting autonomously in an environment to meet its designated objectives [4]. Agents are distinguished by their ability to make decisions independently, perceive their surroundings, and interact with other agents or systems to achieve complex goals. When multiple agents interact within a shared environment, cooperatively or competitively, to solve problems or perform tasks, they form aMAS [1]. MASs use the collective capabilities of individual agents, coordinating their actions and decisions to address problems beyond the scope of any single agent. Collaborative behaviour is essential in many applications, such as automated trading systems, smart grid management, robotics, and smart agriculture. The complexity and constant changes in the environment require an intelligent and distributed approach to decision-making and control. Through the coordinated operation of multi-agent systems, achieving greater efficiency, adaptability, and resilience in both digital and physical systems is possible. This marks a significant advancement in how we approach problem-solving and system design in cyber-physical systems [21, 24].

The integration of MAS and DTs represents a milestone in simulating complex interactions and decision-making processes that mirror real-world dynamics. DTs become more powerful through their functionality and intelligence, enabling real-time optimisation and scenario planning across various scenarios. The application of MAS within DT frameworks represents a significant advance toward automated systems that can predict failures, dynamically adjust to changing conditions, and innovate through continuous learning. This integration is particularly beneficial in domains where complexity and unpredictability play a significant role [6]. It leads the way for a future where digital and real entities connect seamlessly, improving performance and operational insights.

There are numerous articles [17, 34] available that discuss the concept of digital twins in general. However, our primary goal in this article is to identify how multi-agent systems have been used to implement Digital Twin. To achieve this, we defined the research questions and found the answers. Our key contribution to this article is to identify the gaps and future directions in the current research area. We highlight the need for improved scalability, adaptability, and interoperability within the DT and MAS frameworks to meet the evolving demands of complex domains such as manufacturing, smart healthcare, smart cities, and smart agriculture. This article also highlights which domains received more attention and which have more scope to explore this combination. In addition, this article also discusses the proposed architecture for implementing Digital Twins in agriculture.

The remainder of the section is structured as follows: Section 2 introduces the background on DT and multi-agent systems. Section 3 outlines the methodology. In Section 4, we present the findings, including application domains (4.1), agent features (4.2), technologies used in digital twin implementation beyond agents (4.3), and future directions (4.4). Section 5 covers the discussion and future work. Finally, Section 6 concludes the article.

2 Background

2.1 Digital Twin

In recent years, the topic of Digital Twin has gained considerable attention around the world. The concept of DT refers to creating a virtual/digital replica or model of a physical object, which can be a machine, a building, a farm, or a human being. The digital replica or model is created using real-time data collected from the physical object. This allows for in-depth analysis, simulation, and optimisation, offering a new dimension in managing and understanding these entities. The first terminology was introduced by Grieves in a 2003 presentation and later documented in a white paper setting a foundation for the developments of Digital Twins [14].

DT merges the physical and digital worlds to create dynamic [43, 48], self-evolving models [43] that reflect real-life entities ranging from individual components to complex systems. This idea

fundamentally involves developing a virtual or digital counterpart DT [27] for every physical object, be it a part, machinery or devices, a process, or even human beings. These digital instances are not just static representations, but living models that continuously adapt and evolve [11, 27, 28]. They capture the current state of their physical counterparts in real time, supported by a continuous stream of real-time data. The data include both real-time data and historical data, providing a comprehensive view of the subject's lifecycle.

A Digital Twin is special because of its ability not only to mirror the physical entity, which is a representation of the current state but also to influence the physical entities. This two-way interaction signifies that changes in the digital model can have direct implications on the physical object and vice versa. This capability extends from the microatomical level to the macro-geometric level, encompassing all functional features and interconnections within the working elements [43, 57]. By integrating simulations and service data, Digital Twins offers a multi-dimensional view that is continuously updated, enabling improved decision-making processes. These models become invaluable tools in design and operational environments, forecasting current and future conditions to optimise the performance, maintenance, and overall health of the physical system throughout its lifecycle. Digital models are created by collecting real-time data from IoT solutions, such as sensors and other sources. Although digital twin systems simulate predicted futures, their main distinction lies in providing real-time insights. Unlike simulations [26] that may not account for real-time disruptions, digital twins offer fluidity and actionable insights by incorporating up-to-date data.

2.2 Multi-Agents Systems

MAS are a branch of Distributed Artificial Intelligence [10] that involve a group of autonomous agents working together to achieve a common goal. Agents can cooperate or compete, share knowledge with each other, and operate in a loosely connected environment. The key elements of an agent system include the entity, environment, parameters, and action. The entity refers to the type of agent, which can be software, hardware, or a combination of both. The environment refers to the place where the agent is located and can include various features such as accessibility, determinism, dynamism, and continuity. The parameters refer to the different data types that an agent can sense from the environment. The benefits of using MAS include increased efficiency, low cost, flexibility, reliability, scalability, and reusability. The authors of [10] categorise the features of MAS as leadership, decision function, heterogeneity, agreement parameters, delay consideration, topology, data transmission frequency, and mobility.

In recent years, multi-agent systems have become more popular primarily due to their adaptability and efficiency in handling complex, dynamic, and distributed problems. Researchers have been actively exploring the use of MAS across a wide range of domains, including but not limited to robotics, supply chain management, smart grid management, traffic control, and environmental monitoring. Among these, areas such as smart manufacturing [22], healthcare [8], smart city [29], and smart agriculture [19] have gained significant attention, showcasing the potential of agents to improve automation, decision-making processes, and system resilience in digital twin applications.

Multi-agents offer several key features that are crucial in the implementation of Digital Twins in our vision. Firstly, they demonstrate exceptional performance in distributed processing [5], capable of distributing computational tasks across a network of agents, enabling parallel and distributed computation, essential for handling the vast amounts of data and complex simulations often associated with Digital Twins. Second, their scalability [59] allows systems to quickly adjust to the size and complexity of the Digital Twin environment, making them well-suited for managing diverse and dynamic physical systems. Multi-agents also exhibit interoperability [38, 41], as they can seamlessly interact and communicate with various components of the Digital Twin, including

sensors, actuators, and other agents. This interoperability ensures smooth integration and data exchange within the system.

Moreover, multi-agents possess adaptability [3], allowing them to adjust to changing conditions and requirements within the Digital Twin environment. They can modify their behaviours, strategies, and goals to respond to new information or challenges. Their capability for cooperation and coordination [16] allows them to collaborate and synchronise actions, facilitating real-time decision-making among agents in the context of Digital Twins. Furthermore, their decentralised operation [50] reduces reliance on a central controller, improving robustness and fault tolerance. Multi-agents can also dynamically allocate tasks [58] based on their capabilities and availability, optimising resource utilisation and improving efficiency. Some multi-agents can learn from their interactions and experiences, incorporating machine learning techniques to improve decision-making and predictive capabilities.

Lastly, multi-agents excel in handling real-time data [13] generated by sensors and other sources within the Digital Twin. They can efficiently process these data to make timely decisions and adjustments, contributing to the overall functionality of the system and resource optimisation. These features collectively make multi-agents valuable in implementing Digital Twins, improving their adaptability, scalability, and real-time decision-making capabilities.

3 Methodology

As a first step in exploring the integration of agent technology and Digital Twins, we identified potential key questions. These questions focus on the integration of agents in Digital Twins, the features of agents used, the additional technologies used, and the recommended future directions in this rapidly evolving field. The aim is at reviewing the existing works to address these questions in the combination of agent-based Digital Twin implementations and highlight the future direction in this area.

The identified **Research Questions** (**RQs**) and the objectives of those questions are the following,

- RQ1: What domains utilise agents in Digital Twin implementations?
  Objective: To identify and categorise the domains where agent technology is integrated within Digital Twin frameworks.
- RQ2: What features of agents are utilised by authors in Digital Twin applications?
  Objective: To analyse and summarise the key characteristics and functionalities of agents as presented in the research articles, focusing on how these features contribute to the effectiveness of Digital Twin implementations.
- RQ3: What are other technologies integrated with agents in the implementing Digital Twins?
  Objective: To explore the range of technologies combined with agent systems in Digital Twins, highlighting how these integrations improve the capabilities and performance of Digital Twin solutions.
- RQ4: What are the future developments and opportunities for integrating agents in Digital Twins?
  Objective: To identify opportunities and insights for future research and development.

In this review of the literature on the integration of agent technology and Digital Twins, our goal was to identify relevant research articles that specifically demonstrated the applications of Digital Twins integrated with agent technology. To achieve this, we followed a systematic approach. We used the query string (“agent” AND “digital twin”) on Google Scholar to find relevant literature. Google Scholar was chosen to minimise bias against any specific scientific publisher, adhering to the methodology recommended in [51]. The search covered the period from 2020 to 2024 and

was conducted in January 2024. Our inclusion criteria were designed to ensure the relevance and quality of the selected articles. The criteria were as follows:

(1) Relevance to Topic: Articles must demonstrate the application of Digital Twins integrated with agent technology. This criterion was chosen to focus on studies that directly address the research area of interest.
(2) Publication Period: Articles published between 2020 and 2024 were included to capture the most recent advancements and trends in the field.
(3) Language: Only articles written in English were considered to maintain consistency and comprehensibility.
(4) Original Research: We excluded duplicate publications, follow-up extensions by the same authors, and review articles to ensure the originality of the content.

The initial search yielded 16,100 results. We reviewed these results until we encountered ten consecutive irrelevant or duplicate articles, resulting in 62 initially selected articles. From there, we applied our inclusion criteria, which excluded duplicates, follow-up extensions, and non-English articles. At the end of the inclusion criteria, we selected 22 articles using the combination of Digital Twin and multi-agent systems.

4 Findings

The Table 1 presents the articles reviewed according to the criteria outlined in Section 3. This section provides answers to the defined research questions, such as application domains, agent features used in digital twins, the technologies used in addition to multi-agents, and the future direction discussed by the authors.

4.1 Application Domains

To address the first research question, we investigated the application domains of DT implementation, focusing on integrating multi-agent systems. Integrating multi-agent systems across various domains highlights the effectiveness of agent-based approaches in tackling complex and dynamic problems. In manufacturing, research articles such as [54, 56], and [24] emphasise the role of agents in improving quality control, bi-manufacturing processes, and the overall efficiency of manufacturing systems. These studies demonstrate how agents can facilitate real-time monitoring, decision-making, and automation, improving productivity and adaptability in rapidly changing industrial environments. The application of agent-based systems extends beyond traditional manufacturing into areas such as **Cyber-Physical Systems** (**CPS**) and Edge Computing Environments, as explored in [32] and [53]. These contributions highlight the capability of agents in managing complex interactions between physical and digital entities, enabling more advanced control and optimisation strategies that suit the specific demands of CPS and edge computing scenarios.

In the context of precision farming and agriculture, [15, 44] and [18] illustrate using multi-agent systems to improve efficiency and productivity. The authors of [44] have similar concepts in [45] and [46]. These agents can simulate and manage various agricultural processes, suggesting a significant leap toward automation and intelligent resource management in the agricultural domain. Smart cities and healthcare are other critical areas that benefit from agent-based technologies. [6] and [35] discuss the deployment of agents in urban planning and emergency department management, respectively. Agents are instrumental in optimising public health services and improving the quality of urban life by addressing challenges related to traffic, public safety, and emergency medical services. In the specialised field of advanced air mobility, [7] presents a futuristic view where agents contribute to the safety and efficiency of unmanned aircraft systems. This indicates a broader application spectrum, from ground-based manufacturing and farming to the skies above,

Table 1. Digital Twin in Agriculture

Reference	Year	Title	Domain	Purpose of Agents	Other Technologies	Discussed Future work
[56]	2020	A Quality-Oriented Digital Twin Modelling Method for Manufacturing Processes Based on A Multi-Agent Architecture	Manufacturing	Quality control	Semantic web	Ontology development, feasibility study
[44]	2020	Multi-agent approach for developing a digital twin of wheat	Precision Farming	Continuous monitoring and control, yield forecasting and risk management, development of agricultural recommendations	Ontologies and Knowledge Bases, computational simulation models	Self-organisation of agents and domain-specific ontologies of agriculture with machine learning and big data processing methods.
[54]	2021	Bi-level dynamic scheduling architecture based on service unit digital twin agents	Manufacturing	Dynamic scheduling, decomposition and coordination	SCADA, Redis, Kafka, JADE	Scalability and adaptation
[55]	2021	Adaptive digital twin and multiagent deep reinforcement learning for vehicular edge computing and networks	Vehicular edge computing networks	Optimisation of edge computing services, Improve collaboration and efficiency	Deep Reinforcement Learning (DRL)	Advanced learning algorithms, broader application scenarios, integration with 5G and beyond.
[6]	2021	Multi-agent systems and digital twins for smarter cities	Smart Cities	Simulation and optimisation of traffic systems, Real-time data integration and decision-making	MARS Framework, Data processing and integration techniques	Improve model accuracy and predictive capabilities, expansion to other urban systems, interoperability, standardisation.
[33]	2021	Foresighted digital twin for situational agent selection in production control	Manufacturing and Production Control	Situational control agent selection, Improve predictive capabilities	Reinforcement Learning, Monte Carlo methods and Trust Region Policy optimisation (TRPO)	Advanced machine learning integration, broader application across manufacturing domains, interoperability and standardisation
[2]	2021	An agent-based digital twin for exploring localised non-pharmaceutical interventions to control covid-19 pandemic	Public health and Urban planning	Simulation of pandemic dynamics and NPI effectiveness	Data analytics and simulation, computational techniques	Expansion to other pandemics and urban challenges, integration with real-time data, advancements in ABM and simulation technologies
[52]	2021	A digital twin to train deep reinforcement learning agent for smart manufacturing plants: Environment, interfaces and intelligence	Smart Manufacturing Systems	Automating smart manufacturing, Data-driven optimisation	Deep Reinforcement Learning (DRL), Simulation and Virtual Commissioning	Integration with more advanced AI and machine learning techniques, expansion across manufacturing domains
[24]	2021	Digital twin integration in multi-agent cyber physical manufacturing systems	Manufacturing	Interconnectivity and data integrity, operational and decision support	AI, Machine learning	expansion to other domains, improve security measures
[37]	2022	A multi-agent and internet of things framework of digital twin for optimised manufacturing control	Manufacturing	Resource optimisation and scheduling, Improve interaction and data integration	Improved Contract Net Protocol (ICNP), Self-organising and Self-adaptive models	Scalability and flexibility, advanced learning and optimisation techniques
[12]	2022	Autonomous Digital Twin of Enterprise: Method and Toolset for Knowledge-Based Multi-Agent Adaptive Management of Tasks and Resources in Real Time	Enterprise Resource Management	Adaptive resource management, Knowledge-Based decision making	Ontologies, Knowledge-Based toolsets	Expansion other domains, integration with advanced technologies, improvement in ontological models and MAS
[39]	2023	Evolutionary digital twin model with an agent-based discrete-event simulation method	Manufacturing and Industrial Systems	Simulation and optimisation, adaptive learning and evolution	Reinforcement Learning, Agent-Based Discrete-Event (ABDE) Simulation	Hybrid modeling approaches, unsupervised learning techniques, distributed integration methods
[53]	2023	Digital twin-driven collaborative scheduling for heterogeneous task and edge-end resource via multi-agent deep reinforcement learning	Edge Computing Environments	Collaborative scheduling, optimisation of resource allocation	Deep Reinforcement Learning (DRL)	Scalability to other domains, advanced AI and learning algorithms

(Continued)

Table 1. Continued

Reference	Year	Title	Domain	Purpose of Agents	Other Technologies	Discussed Future work
[30]	2023	Madtwin: a framework for multi-agent digital twin development: smart warehouse case study	Smart Warehouses	Autonomous and Cooperative action, Improve operational efficiency	JADE MAS Platform	Scalability to other domain, advanced integration with IoT and AI
[35]	2023	An Agent-Based Architecture of the Digital Twin for an Emergency Department	Healthcare, specifically Emergency Departments (EDs)	Simulation and optimisation, Decision support	Agent-Based Simulation (ABS, Monte Carlo Simulations	Expansion to other healthcare areas, real-Time data integration, improve predictive analytics
[36]	2023	A multi-agent and cloud-edge orchestration framework of digital twin for distributed production control	Distributed manufacturing	Efficient resource allocation and scheduling, real-time decision making and execution	Deep learning algorithms	Advanced machine learning and AI integration, expansion across manufacturing sectors
[7]	2023	Developing a Digital Twin for Testing Multi-Agent Systems in Advanced Air Mobility: A Case Study of Cranfield University and Airport	Advanced Air Mobility and Unmanned Aircraft Systems	Simulation and testing, rapid prototyping	Microsoft AirSim and Unreal Engine, Cesium for Geospatial visualisation, Python-based architecture	Improve mixed-reality and HIL testing, broader application across AAM and UAS Ecosystems, integration of advanced AI and machine learning algorithms
[23]	2023	Digital twin-enabled automated anomaly detection and bottleneck identification in complex manufacturing systems using a multi-agent approach	Manufacturing	Automated anomaly detection and bottleneck identification, Decision-making and control	Cloud, Big data processing	Expansion to other industrial sectors, integration of advanced AI and machine learning
[15]	2023	Towards Agrirobot Digital Twins: Agri-RO5—A Multi-Agent Architecture for Dynamic Fleet Simulation	Agriculture and Agrirobotics	Dynamic fleet simulation, realistic testing and validation, Dynamic multi-robot task allocation	Robot operating system (ROS), Unity3D and SPADE3 middleware, FIVE framework	Broadened application spectrum, integration with advanced technologies
[32]	2023	Towards developing digital twin enabled multi-agent systems	Cyber-Physical Systems (CPS)	Improving interconnectivity and data integrity, operational efficiency and decision support	JaCaMo and WLDT	Dynamic and scalable integration
[18]	2023	Digital twin deployment for smart agriculture in Cloud-Fog-Edge infrastructure	Smart Agriculture	Decision making	Cloud-Fog-Edge, Semantic web	More interoperability, improve decision making
[25]	2024	A multi-agent digital twin–enabled decision support system for sustainable and resilient supplier management	Supplier management within the manufacturing sector	Decision support system	Stratified fuzzy decision making, SAP system integration	Integration with external platforms, advanced simulation techniques

showcasing the adaptability and potential of agent-based systems. Finally, in the domain of supplier management within the manufacturing sector, [25] highlights how agents can revolutionise the approach to managing complex supplier networks, ensuring resilience, sustainability, and efficiency. This highlights the critical role of agents in improving operational aspects and contributing to strategic and sustainable business practices.

In the analysis of 22 research articles, it is observed that agent-based systems were most prominently utilised in the manufacturing domain, as depicted in Figure 1, with ten articles demonstrating their critical role in this field. This highlights the importance of agents in DTs for industrial development. In addition, other sectors, such as agriculture, healthcare, and smart cities, have significant potential to explore using agents effectively in the implementation of DT. This expansion could lead to significant advancements in sustainability, efficiency, adaptability, automation, and quality of life.

The reviewed articles show that integrating DTs and multi-agent systems across diverse domains such as manufacturing, smart cities, healthcare, and agriculture holds significant potential. It demonstrates how these technologies can potentially transform these fields to the next level.

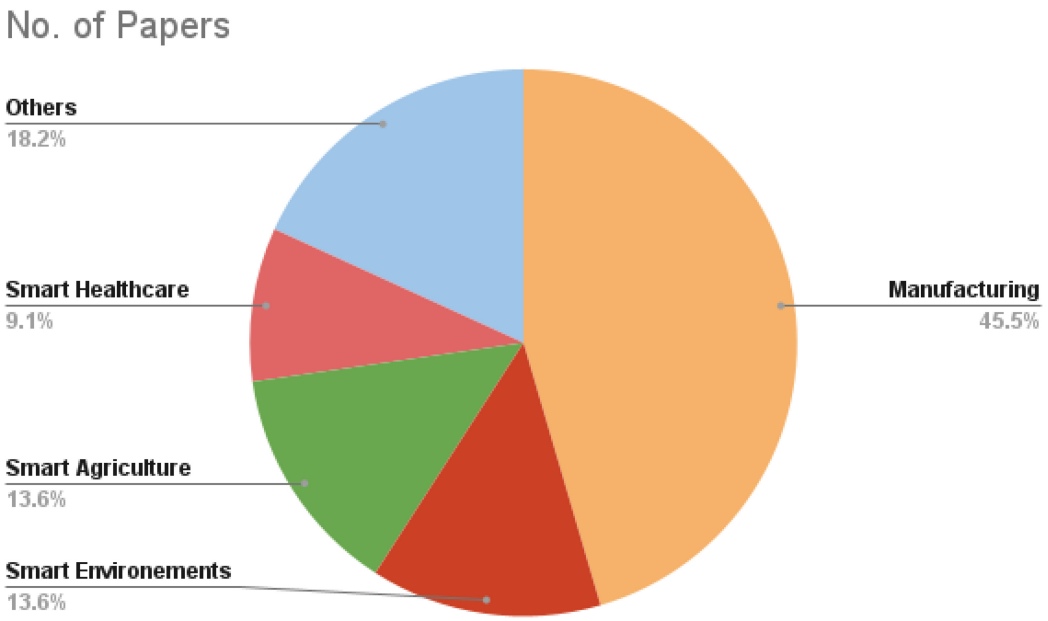


Fig. 1. Number of reviewed articles in various domains.

Agents play a crucial role in implementing DTs and improving the system by enabling more dynamic, sophisticated, and autonomous decision-making processes.

4.2 Agent Features

The use of agents in DT implementations across various studies demonstrates the diversity and depth of applications that improve decision-making, operational efficiency, and simulation capabilities in numerous domains. Notably, as identified in the reviewed articles, the primary purposes of using agents cover a wide range of functionalities, from autonomous operation and cooperative behaviour to dynamic scheduling and improved interconnectivity.

Several review articles, including [54, 56], and [32], highlight the importance of involving the agent in enabling autonomous and collaborative decision-making within DT systems. This functionality is important for enabling systems to manage resources and schedules adaptively, reflecting a significant move towards autonomous systems capable of self-management and real-time optimisation. For example, [54] emphasises the importance of dynamic scheduling and the decomposition and coordination of tasks, highlighting the ability of agents to navigate complex operational environments through intelligent scheduling and resource allocation.

Operational efficiency and decision support emerge as another purpose for integrating agents within DTs, as mentioned in [18, 32] and [24]. These studies highlight the significant role of agents in improving interconnectivity and data integrity. These aspects are important for ensuring the reliability and effectiveness of DT simulations and their corresponding physical systems. The focus on operational and decision support, particularly in applications like smart manufacturing [52] and advanced air mobility [7], reflects the growing need for systems that not only simulate but also predict and optimise operational outcomes in real time.

Simulation and optimisation, particularly in the contexts of agricultural recommendations [44] and pandemic dynamics [2], show a broad application range of agents in areas requiring sophisticated predictive analytics and scenario exploration. This capability is important for developing

plans and actions in dynamic and uncertain environments. The analysis also identifies collaborative scheduling and resource optimisation, as mentioned in [53] and [36], as critical areas where agents contribute significantly to DT architectures. This functionality is important for systems that require coordination among multiple entities, such as fleet management in agriculture [15] and traffic systems in urban planning [6].

The most common purpose across the reviewed research articles for applying agents within DT implementations is improving decision-making and optimising performance. Agents are important in enabling systems to autonomously adapt, optimise, and make decisions based on real-time data and simulations. Integrating agents with DTs allows dynamic and efficient system management while providing opportunities for AI and machine learning integration, expanding the scope of DT applications across various domains. This reflects a significant trend toward creating more intelligent, autonomous, and interconnected systems capable of addressing complex real-world challenges.

4.3 Technologies Used in Digital Twin Implementation Beyond Agents

From the reviewed articles, it is interesting to note that in integrating agents and DTs, authors have also used various other technologies to address specific challenges and improve the system's capabilities. For instance, **Deep Reinforcement Learning** (**DRL**) has become a critical technology for enabling adaptive decision-making and optimisation within complex environments, as demonstrated in the works of [52, 53, 55]. These researchers used DRL to improve the autonomous capabilities of agents, allowing for more efficient and responsive system behaviours in manufacturing and edge computing contexts.

Agent-Based Modeling (**ABM**) and Simulation, including **Agent-Based Discrete-Event** (**ABDE**) Simulation, are prominently featured in the methodologies of [39] and [35]. These techniques provide a dynamic and granular approach to replicating complex interactions within DTs, offering valuable insights into emergent behaviours and potential system optimisations. The integration of ontologies and knowledge bases by [12, 44] and [18] highlights the importance of semantic clarity and data interpretability in improving the decision-making processes of multi-agent systems. Through structured data and domain-specific knowledge, agents can make more informed and accurate decisions.

Furthermore, the adoption of comprehensive simulation platforms such as Microsoft AirSim and Unreal Engine, as well as Cesium for Geospatial Visualisation by [7], facilitates the creation of highly realistic and immersive simulation environments. These platforms are critical to achieving geospatial accuracy and detailed environmental visualisation, essential for effectively simulating and testing agent behaviours in advanced air mobility scenarios.

In summary, the reviewed articles highlight a trend towards using a combination of simulation technologies, advanced machine learning algorithms, and structured knowledge representation techniques to improve the capabilities of DTs and multi-agent systems. These technologies not only facilitate the simulation of complex scenarios and the optimisation of system operations but also enable the systems to adapt to changing conditions through learning and decision-making processes. Combining these technologies within the DT framework represents a significant advancement. This contributes to developing intelligent/cognitive, flexible, adaptable, and highly efficient DT systems designed to address the complex challenges in modern industrial and operational environments.

4.4 Future Direction

To answer our research question 4, we analysed the reviewed articles to identify critical gaps or future works discussed by authors in integrating DT and MAS. Studies [23] and [15] have highlighted

the urgent need for improved scalability and adaptation within these systems. This highlights a fundamental need for improvement in the ability of current DT and MAS configurations to dynamically adjust and scale in response to the complexities of evolving operational environments. Research works [2] and [7] emphasise the critical role of advanced technologies such as AI, machine learning, and the **Internet of Things** (**IoTs**) in pushing the boundaries of what DTs and MAS can achieve. The gap identified in the existing frameworks' capacity for intelligent and autonomous operation suggests that deeper integration of these cutting-edge technologies could substantially improve aspects such as predictive analytics, decision-making, and overall system efficiency.

The authors of [36] and [53] also indicate a significant research gap in the adaptability of DT and MAS applications. The potential of DT and MAS to revolutionise sectors beyond their conventional uses, such as healthcare, urban planning, and agriculture, highlights the need for research to expand applicability and bridge existing gaps. Moreover, the call for improved interoperability and standardisation, as discussed in [18, 30] and [35], highlights prevalent challenges in ensuring seamless integration and functionality of DT systems across varied platforms and industries. Addressing this research gap is important to promoting wider adoption and innovation within the DT and MAS ecosystem.

Overall, the common features discussed by authors as future work are scalable, adaptable, and seamlessly integrated with advanced technological capabilities. Other key things discussed by the authors are to fully realise the potential of DT and MAS in addressing the complex challenges of today's industries and society; it is essential to broaden their application range, improve their ability to work together and develop standardised methods.

5 Discussion and Future work

This review is inspired by the visionary article by Mariani, S et al. [31]. The authors investigate the integration of Digital Twins, agents, and Multi-Agent Systems in improving complex socio-technical and cyber-physical systems. Their research highlights the potential of agent-oriented interpretations to enrich DT capabilities and vice versa, emphasising cross-fertilisation as a path to improving system functionalities. They discuss challenges related to interoperability and accurate digital mirroring of the physical world, highlighting the integration of advanced technologies like AI and IoT as avenues for advancing intelligent system functionalities such as predictive maintenance and anticipatory planning. While Mariani et al. provide an overview of the potential collaborations between DT and agent technologies, their analysis primarily stays at a strategic level. They touch on the intersection of these technologies but explore less into the detailed applications of agents within DT frameworks across various sectors or the specific functions these agents may offer.

Contrastingly, our review deeply explores the practical and theoretical applications of agents within DT implementations across various domains, including manufacturing, CPS, smart cities, healthcare, advanced air mobility, and agriculture by analysing the existing works. We analyse agent functionalities such as autonomy, decision-making, learning, and adaptability and their critical roles in improving DT implementations. By examining 22 research articles, our study not only identifies the transformative potential of MASs in manufacturing but also articulates significant opportunities to explore agent technology in other sectors to improve sustainability and efficiency. Furthermore, we highlight the increasing role of complementary technologies such as Deep Reinforcement Learning, Agent-Based Modeling, Ontologies, and advanced simulation platforms, detailing how these integrations address specific challenges and improve system capabilities. Our work concludes by identifying critical research gaps and future directions, emphasising the need for scalability, adaptability, improved interoperability, and the extension of DT and MAS applications beyond conventional uses.

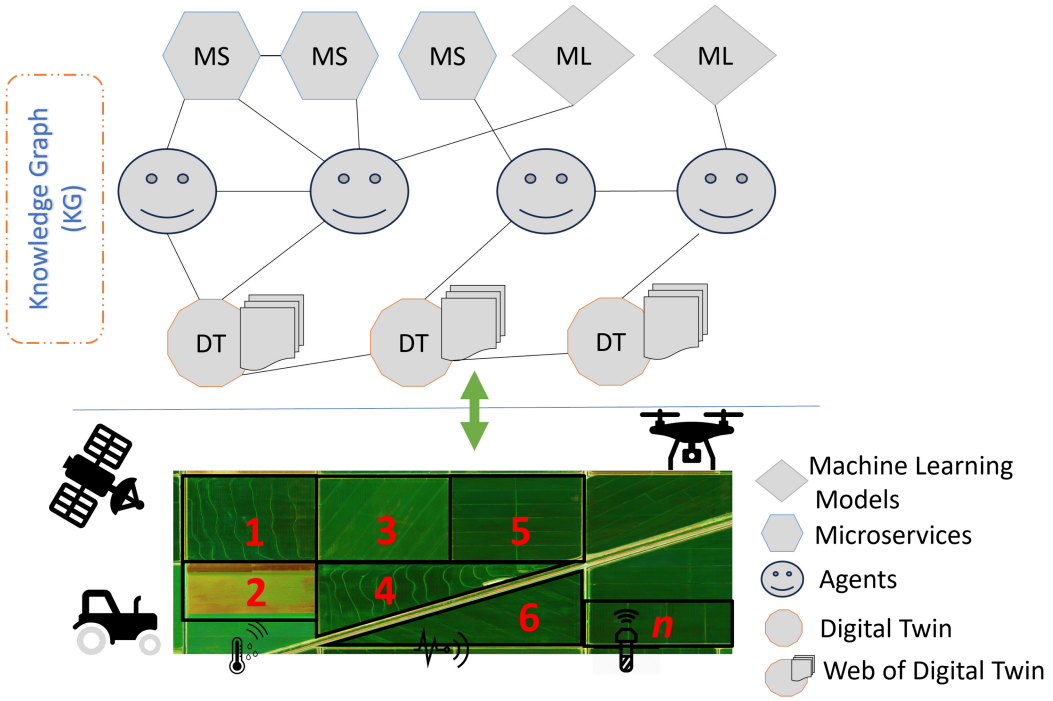


Fig. 2. Proposed architecture for a digital twin in open-environment arable farming.

Currently, we are in the process of implementing a web of digital twins for open-environment arable farming within the smart agriculture domain. The early prototype [18, 20] has demonstrated the feasibility of implementing Digital Twins using agents. The agents were primarily used for monitoring, prediction, decision-making, and adaptation. This literature review has guided our research towards integrating cutting-edge technologies such as semantic web technologies and optimising the effectiveness of agent features within our research framework. Our primary objective for future work is to improve the implementation of DTs by incorporating insights from the comprehensive reviews conducted during this study. By developing a DT system for open-environment arable farming that incorporates multi-agent systems, we anticipate improvements in interoperability, adaptive and predictive capabilities, intelligent decision-making, and personalised, context-aware services. Moreover, semantic web technologies have the potential to improve the accessibility and precision of data for decision-making, enabling farmers and producers to make accurate decisions. By expanding the limits of current technology, we aim at using technological advancements in agriculture to sustain the world's growing population.

Based on our review and conclusions, we propose a high-level architecture, illustrated in Figure 2. This architecture includes a **user interface** (**UI**) as the central point for user input, allowing users to input critical parameters such as farm and field identifiers, sowing dates, and other essential data to initiate the Digital Twin. The UI acts as a gateway for users to interact with the DT system, feeding the entered data into a network of microservices such as weather forecasting, soil analysis, crop growth modelling, irrigation scheduling, and agents. These agents operate autonomously, performing tasks such as data collection, activating microservices, or making decisions based on the outputs from **machine learning** (**ML**) models. Examples of these agents include manager, farm, field, and recommendation agents. In this context, a farm is viewed as a

collection of fields, leading to the creation of an individual DT for each farm and field, thus forming a web of digital twins [40].

The system uses machine learning models to predict crop yield, detect plant diseases, optimise resource allocation, and more. These models continuously learn from field-gathered data, improving their accuracy over time. A knowledge graph, updated with new findings by the agents, supports querying for knowledge and user notification about significant changes. The UI is instrumental in collecting data and displaying actionable insights through "what-if" scenarios and real-time updates from the DT of each field. This configuration offers a comprehensive and interactive experience for managing open-environment arable farms, equipping users with a potent tool for analysing and predicting optimal farming strategies.

Agriculture is a complex and dynamic domain characterised by uncertainty in environmental conditions. Effective agricultural planning requires foresight, with plans needing to be made well in advance. However, executing these plans demands significant resource investment, often involving substantial costs. Given the dynamic nature of agriculture, these plans must remain adaptable to accommodate real-time environmental changes. Additionally, the proposed architecture can be applied to any domain that shares similar attributes to agriculture.

The combination of agents and Digital Twins offers significant benefits, but several important questions must be addressed to fully realise the potential of this integration in the future. Implementing DT with multi-agent systems can be resource-intensive, so examining the economic feasibility and resource demands of such implementations is crucial. Additionally, ensuring seamless interoperability between different systems and technologies is a key challenge that needs to be addressed. Defining and measuring the performance and effectiveness of agents within DT implementations is another area that requires careful consideration. Furthermore, ethical and privacy concerns, including data security and user consent, must be thoroughly examined. It is also important to incorporate diverse perspectives, drawing insights from various industries, stakeholders, and academic disciplines. Finally, achieving and maintaining real-time data synchronisation and accuracy in dynamic environments presents a significant challenge that must be overcome.

6 Conclusion

In conclusion, DT technology has emerged as a significant innovation, focusing on adaptability, data integration, modelling, simulation, automation, prediction, and real-time data management. This article has investigated the role that agents play within DT implementations, highlighting their functions in achieving these features. Our analysis revealed that multi-agent systems are integral to DT, serving diverse purposes such as decision support, simulation, automation, and prediction.

Our findings show that the manufacturing sector dominates research in the DT domain. In contrast, domains such as healthcare and agriculture present potential for expansion and development. Moreover, we have explored different technologies integrated into DT implementations, revealing the complex and diverse nature of DT. It has the potential to integrate various technologies. There are many articles that highlight future expansion opportunities. They suggest the potential for broader applications across domains, incorporation of advanced technologies, scalability, adaptability, improved dynamics, and improved model accuracy.

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